

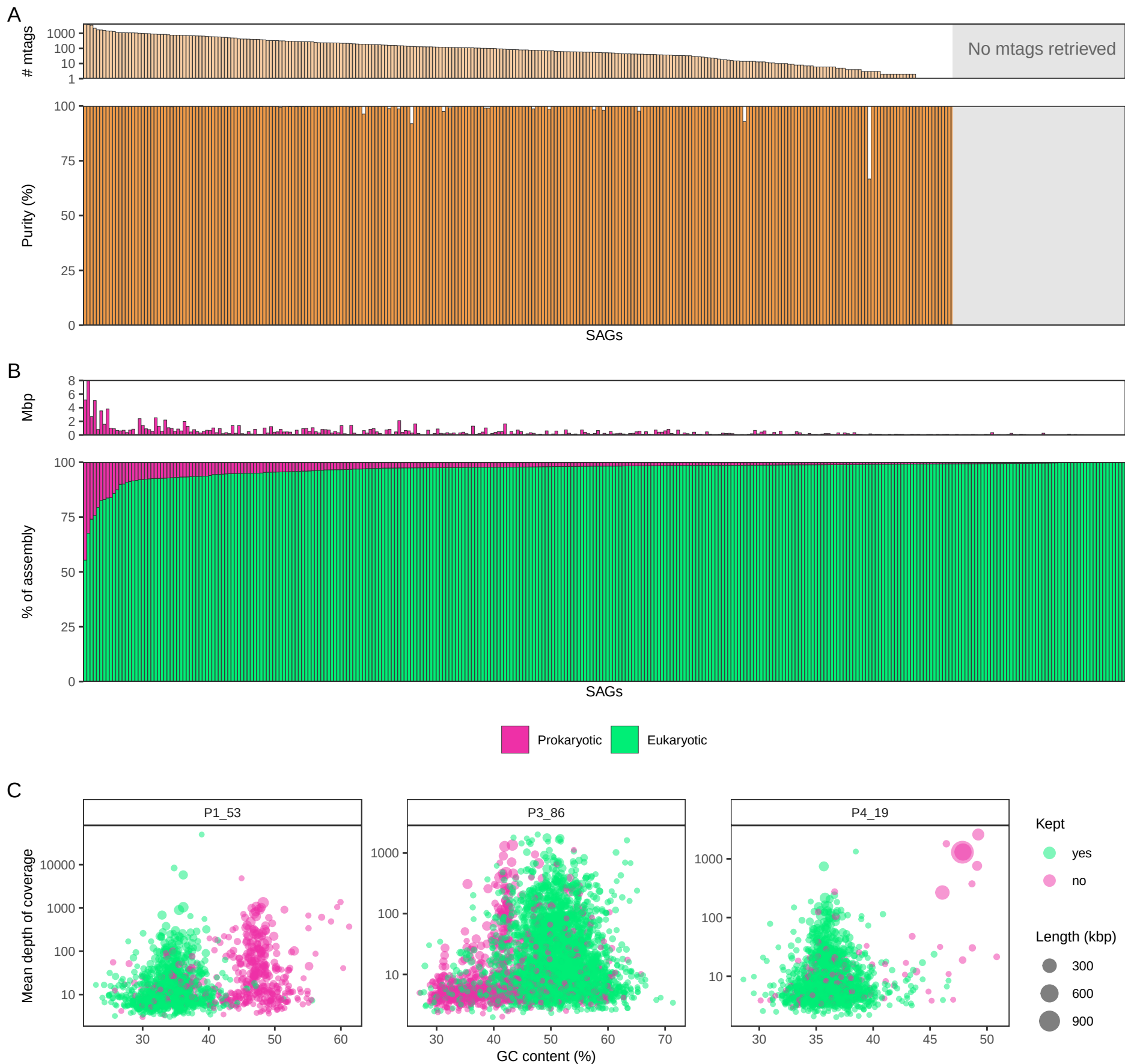
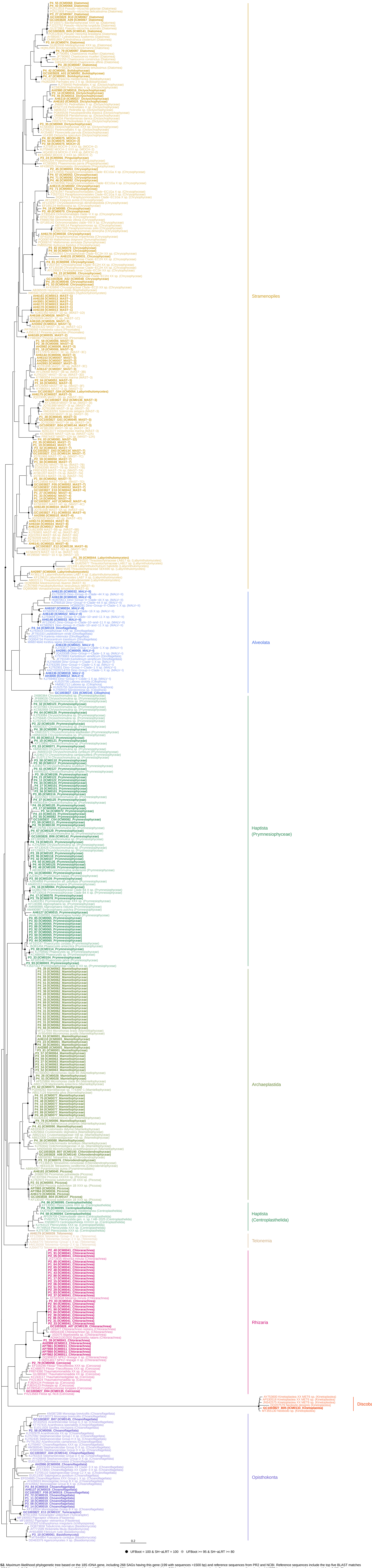


Fig. S1. Analysis of cross-eukaryotic and prokaryotic contamination in the 325 SAGs. (A) Number of mtags (Illumina V4–18S rDNA reads) retrieved in 272 SAGs (upper) and percentage of them affiliating to the same taxonomic group (lower). (B) Assembly size corresponding to prokaryotes in all the SAGs (above) and the percentage they represent (prokaryotic signal in purple and eukaryotic signal in green) in initial assemblies (below). (C) Blob tools analyses of the SAGs with highest prokaryotic signal showing that prokaryotic contigs (in purple) consistently display distinct genomic signatures.



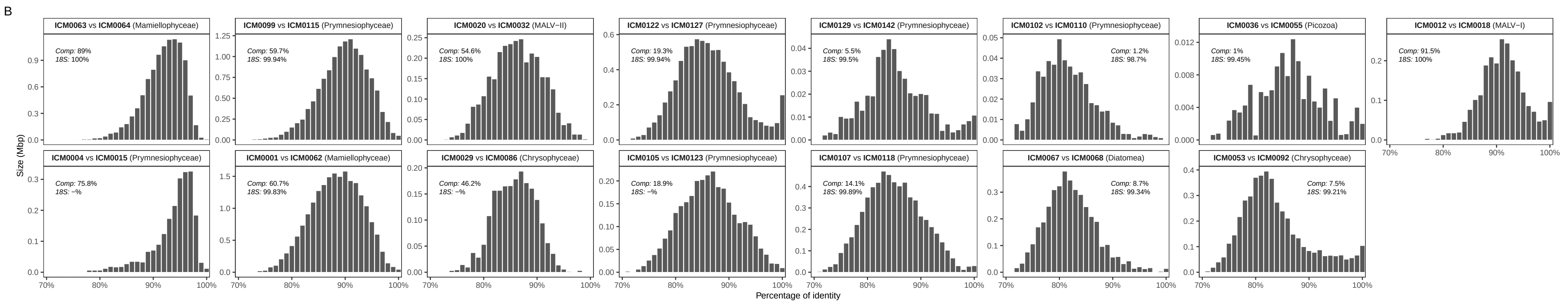
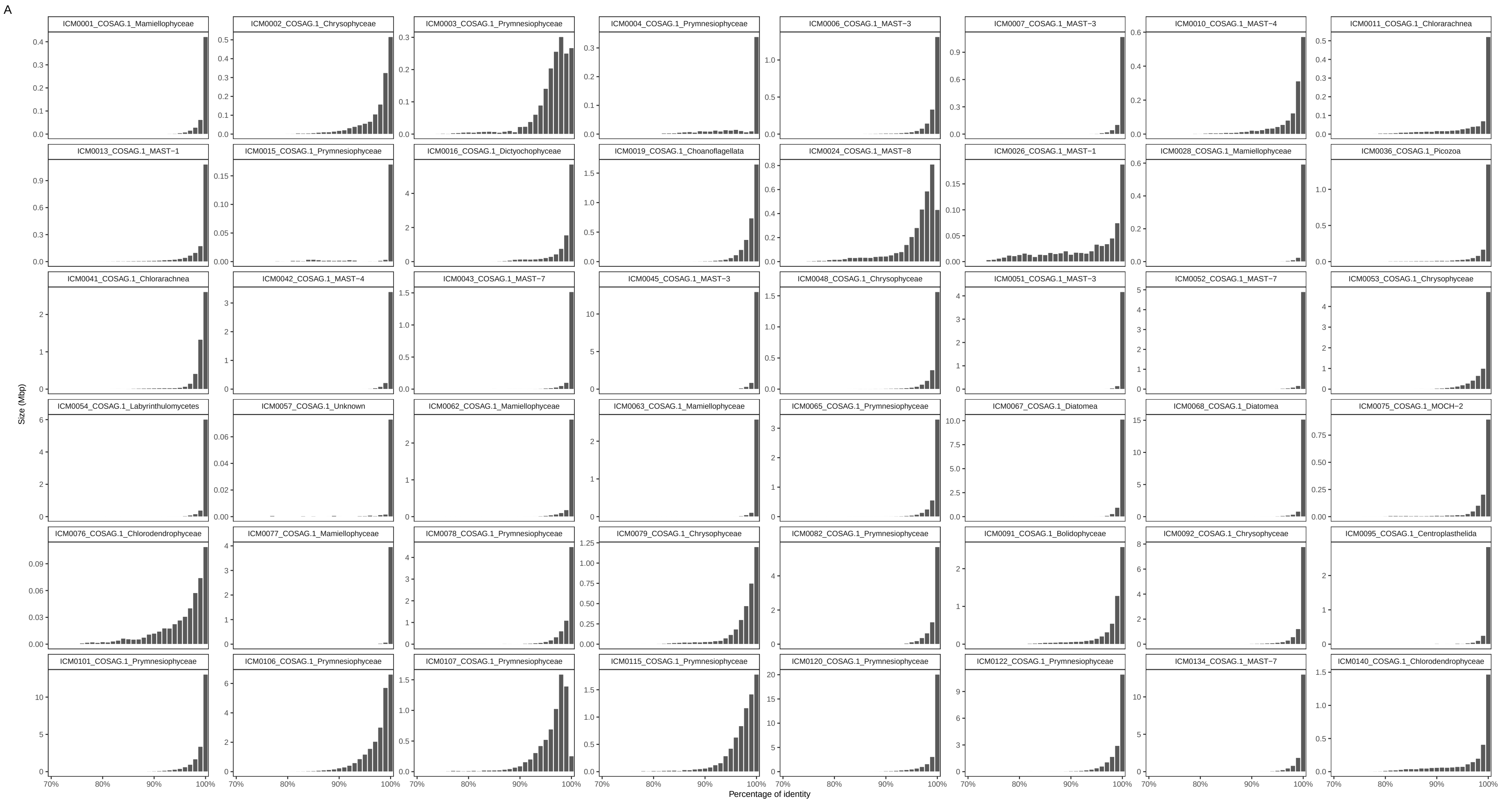


Fig. S3. Genome similarity among SAGs within each co-assembly (A) and between closely distinct genomes (B). Pairwise average nucleotide identity (ANI) comparisons were calculated by fragmenting the subject genome into 1 kb segments and using BLAST hits with a query ≥ 0.5 kb. The similarity against the queried genome size was distributed in 1% bins histograms. Panels from (B), comparing different genomes, also display the sequence similarity of the 18S rDNA genes (the first 7 cases are those matching the same ASV) and the fraction of the genome that has been compared.

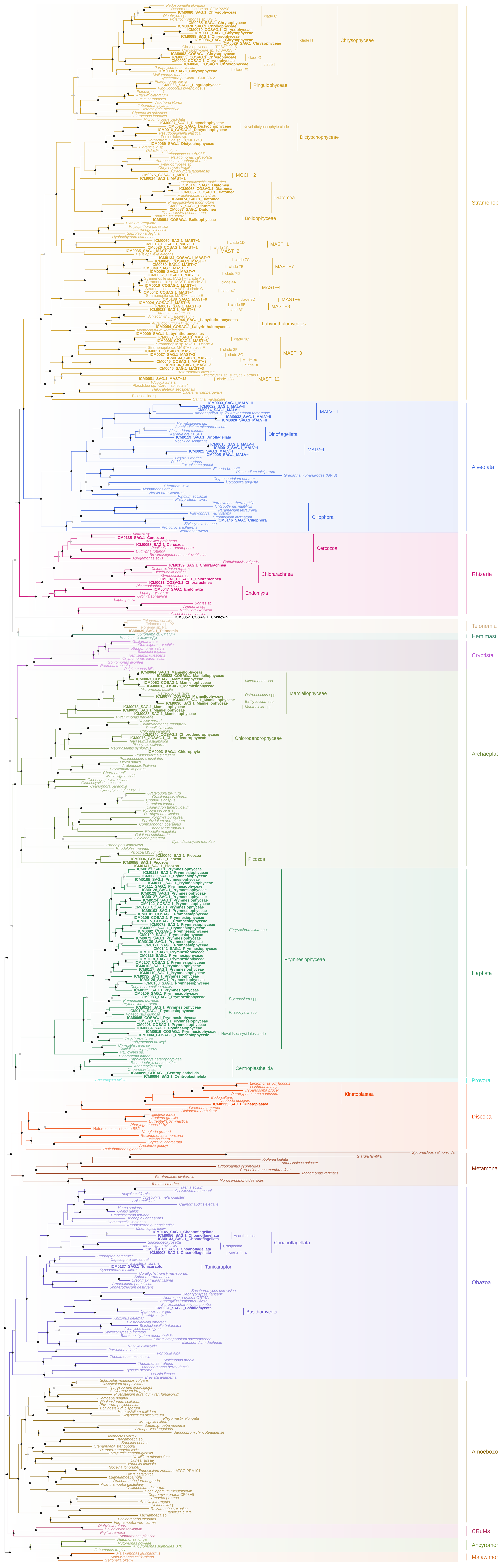
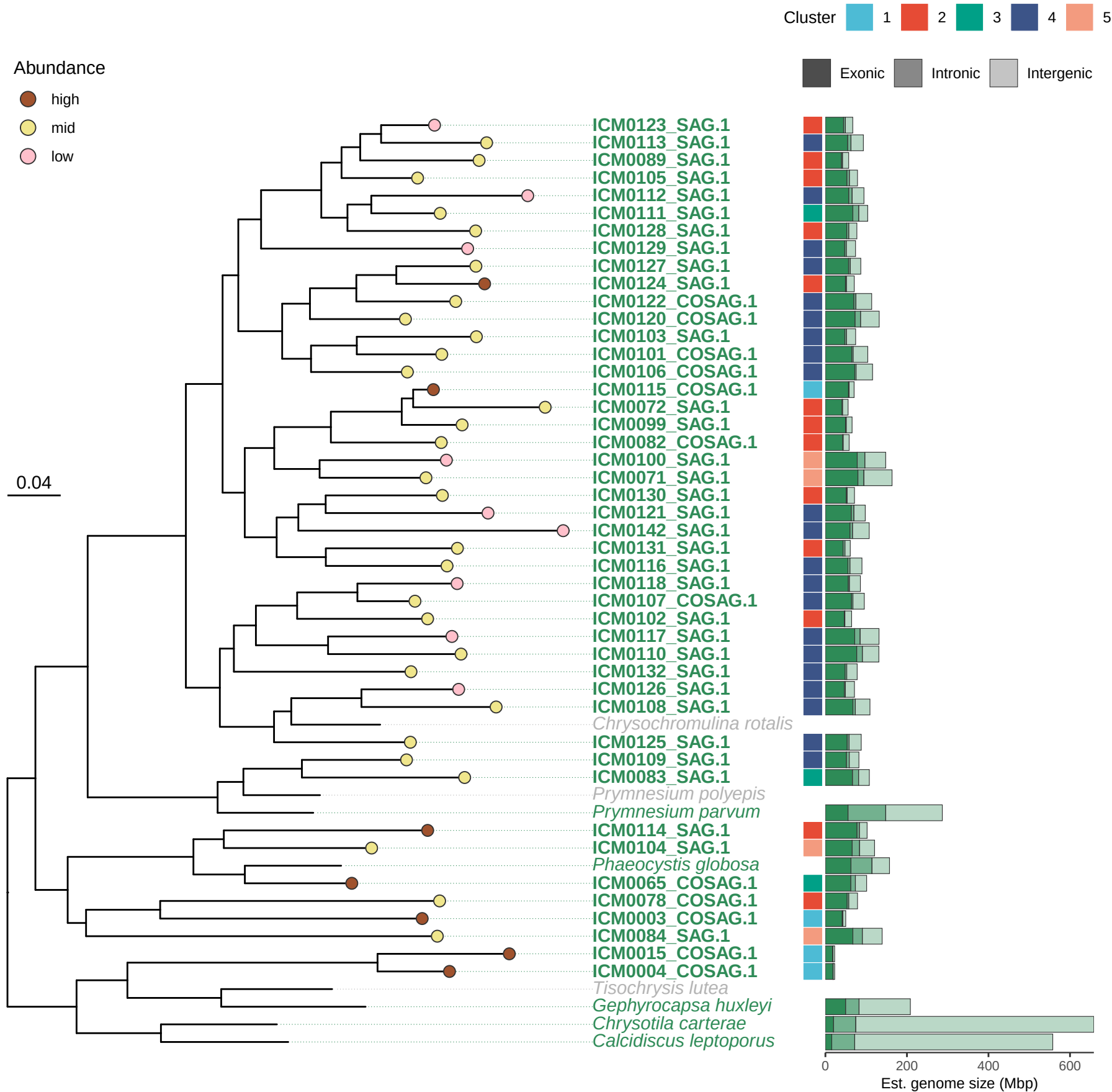


Fig. S4. Full multigene phylogenetic tree of the 147 fungal genomes and 304 reference genomes. The tree was estimated in IQ-TREE with model LG+C60+H+G+PMSF, using a supermatrix of 240 conserved marker proteins constructed with PhyloFisher. SAGs and reference genomes are colored by supergroup. Taxonomic groups containing SAGs are indicated. Bootstrap support is shown for all clades, with nodes highlighted in black when support is 100% and in grey when only slightly lower.

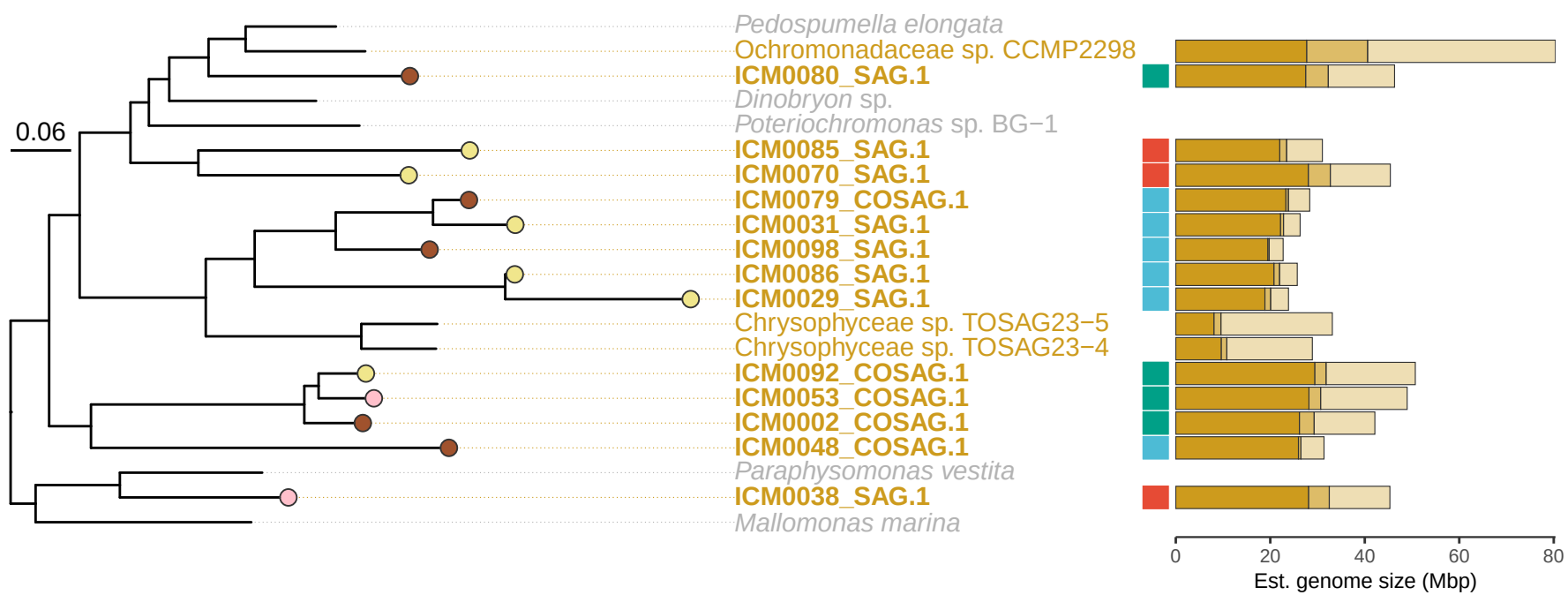


Fig. S5. Biogeography and habitat coverage of ASVs assigned to the final genomes. Distribution of the relative abundance per sample of ASVs corresponding to the final genomes across three large-scale metabarcoding datasets: BBMO (32 samples), OSD (319 samples), and EukBank (8435 samples). The EukBank dataset is further subdivided into five major habitat categories: marine water (6545 samples), marine sediment (634 samples), freshwater (237 samples), freshwater sediment (153 samples) and soils (866 samples).

A



B



C

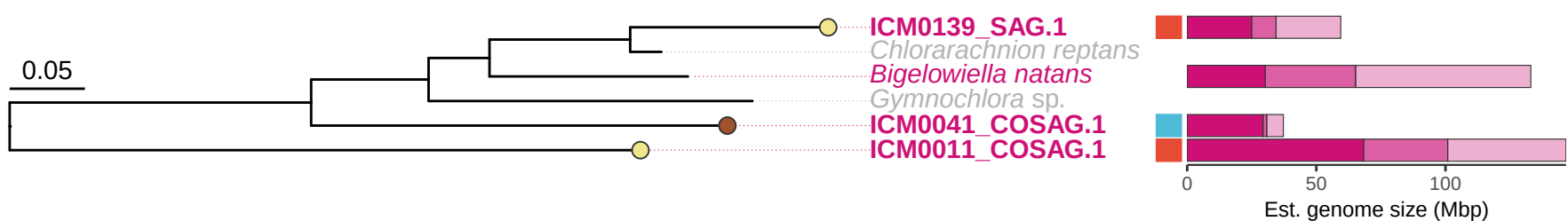


Fig. S6. Genome streamlining in dominant marine species within the available genomic context for Prymnesiophyceae (A), Chrysophyceae (B) and Chlorarachnea (C). A subset of the phylogenomic tree in Fig. S4 is displayed for each group (species in grey lack genomic data). Species represented by SAGs include their relative abundance (colored dots following the code in Fig. 4). When available, the estimated genome size and the distribution among exonic, intronic, and intergenic regions are shown. Many of the most abundant species exhibit indications of genome streamlining, having more compact genomes (smaller and with fewer introns) than closely related species. This pattern is particularly evident for ICM0004 and ICM0015 (A), ICM0079 and ICM0098 (B) and ICM0041 (C).

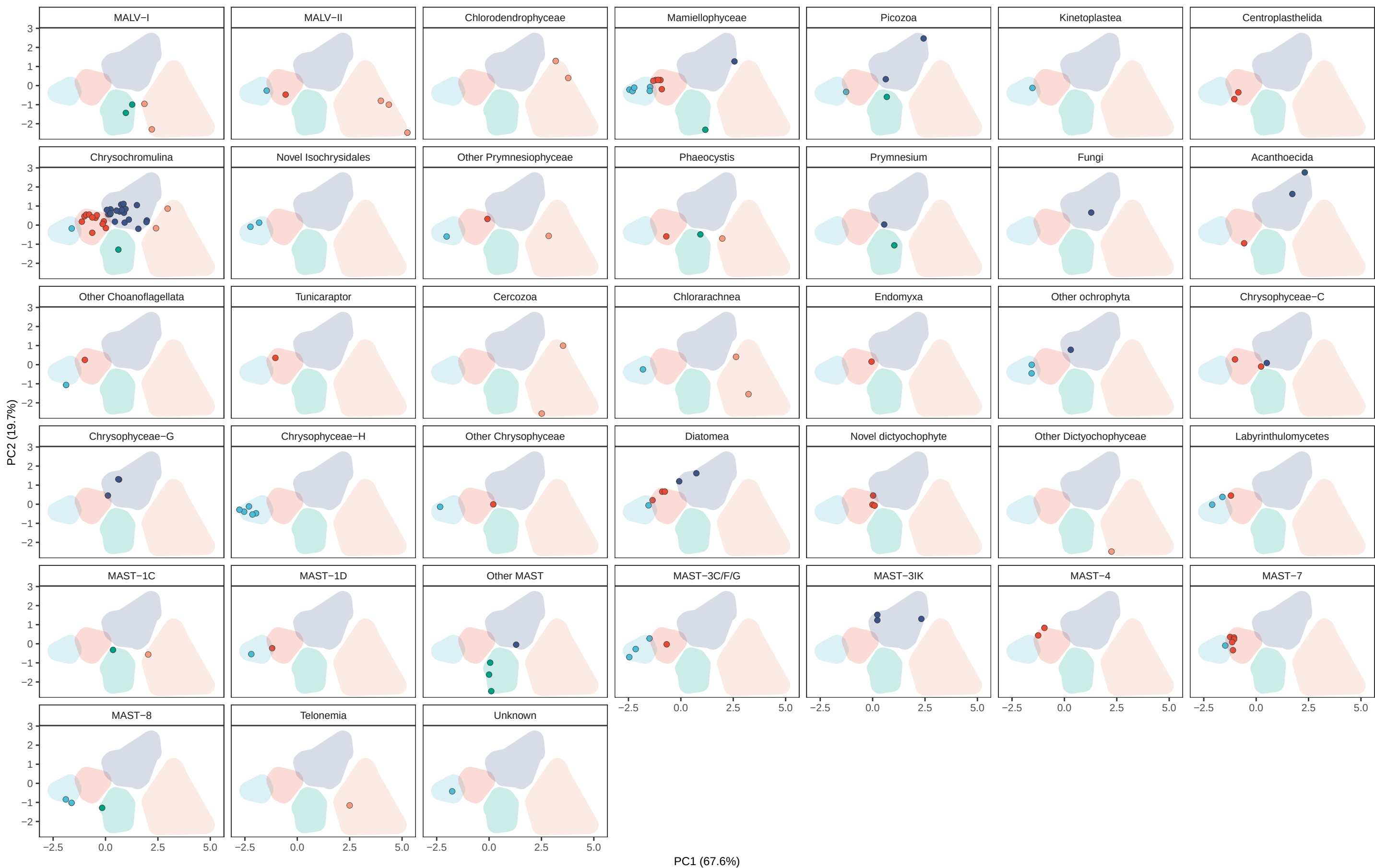


Fig. S7. Placement of SAGs within taxonomic groups in the PCA plot shown in Fig. 4A. Separate plots are presented for each distinct taxonomic group. Shaded areas indicate k-means clusters.



Fig. S8. Pfam domains enriched in SAGs as compared as the EukProt database of protist proteins. They are separated by being enriched only in HP, only in PP, or in the complete pool. Color intensity indicate the percentage of SAGs with the domain enriched in each subgroup.



● UFBoot = 100 & SH-aLRT = 100 ● UFBoot >= 95 & SH-aLRT >= 80

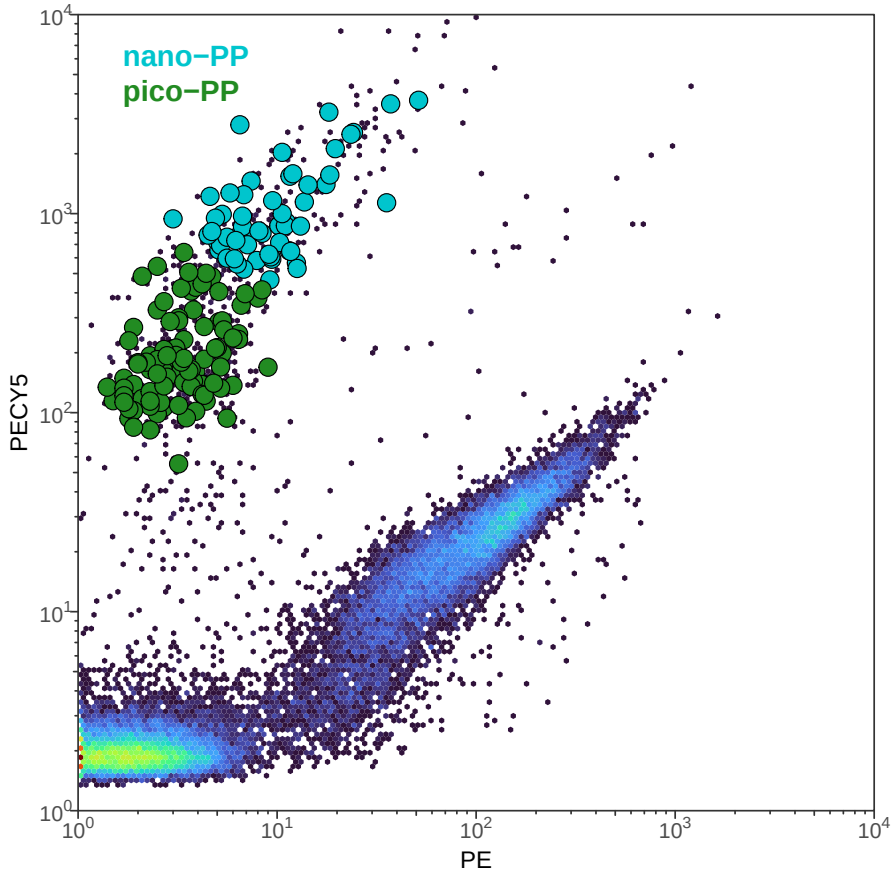


Fig. S10. Flow cytogram of the unstained sample collected in September 2020. Large dots indicate sorted cells, which were classified as pico-PP and nano-PP based on flow cytometric indices.